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An IFC-based framework for semantic integration of BIM and mobile crowd sensing in real-time evacuation routing

Mohsen Rashidian *, Mohammad Reza Malek

Ubiquitous & Mobile GIS Research Lab, Faculty of Geodesy& Geomatics Engineering, K. N. Toosi University of Technology, Tehran, Iran

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ABSTRACT

Emergency evacuation in complex buildings benefits from tight coupling between digital building representations and real-time operational data. While Building Information Modeling (BIM) provides semantically rich indoor descriptions, many evacuation systems still treat sensing technologies and BIM as isolated components, resulting in static or weakly adaptive routing behavior. This paper proposes an IFC-based integration framework that establishes semantic interoperability between BIM entities and mobile crowd sensing (MCS) data to enable real-time, context-aware evacuation routing. IFC models are transformed into a dynamic decision graph in which spaces, exits, and sensors are continuously annotated with live crowd-sensed information. Occupant positions are estimated using Wi-Fi–based indoor localization supported by a two-stage crowdsensing strategy that accounts for both active and silent users. Evacuation routing is formulated as a multi-criteria shortest-path problem using an extended Dijkstra algorithm with dynamically updated weights reflecting distance to hazards, travel time, exit availability, and a route safety index derived from environmental conditions and crowd density. Weight prioritization is formalized using the Analytic Hierarchy Process and examined through sensitivity analysis. The proposed framework is implemented and evaluated in a four-story academic building under three emergency scenarios. Experimental results demonstrate improved evacuation effectiveness, reflected by higher in-time evacuation rates and reduced evacuation times under scenario-specific time constraints. In addition, evacuation times exhibit lower variability under constrained conditions, indicating stable routing behavior when egress options are limited. Overall, the findings confirm that semantic coupling of BIM and mobile crowd sensing enables digital-twin–like behavior for real-time emergency management and illustrates how engineering informatics can transform static building models into operational decision-support systems.

1. Introduction

Real-time decision-making in safety–critical buildings increasingly relies on the integration of semantically rich digital building models with volatile, heterogeneous sensing data. From an engineering informatics perspective, a core challenge is achieving semantic interoperability between Building Information Modeling (BIM) representations and real-time sensing streams so that computational models and routing algorithms can adapt continuously to evolving indoor conditions. Emergency evacuation provides a particularly demanding context: route feasibility depends jointly on building topology, hazard dynamics, and crowd behavior, and must be recomputed under strict temporal constraints.

Mobile Crowd Sensing (MCS) has emerged as a scalable paradigm for collecting occupant-level information during emergencies. By exploiting the sensing and communication capabilities of smartphones and wearable devices, MCS enables decentralized acquisition of location and contextual data and supports dynamic situational awareness [1–4]. Existing work has shown that MCS can reveal population flows, identify bottlenecks, and improve responsiveness in evacuation scenarios. Nevertheless, MCS data streams are heterogeneous, noisy, and largely decoupled from the architectural context in which occupants move, limiting their utility for enforcing building-aware route feasibility. In parallel, BIM has become a dominant paradigm for representing the physical and functional characteristics of buildings [5]. It enables information-rich modeling of indoor spaces and supports advanced analysis for design, construction, and facility operations. In emergency contexts, BIM can represent critical components such as rooms,

* Corresponding author at: Geodesy and Geomatics Engineering, K.N.Toosi University of Technology, Valiasr, Mirdamad Blv. Cross, Post Code, P. O. Box 15875-4416, 19967-15433 Tehran, Iran.
E-mail address: m.rashidian@email.kntu.ac.ir (M. Rashidian).

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corridors, exits, staircases, and fixed sensors, and can support simulation, visualization, and spatial reasoning [6–9]. The Industry Foundation Classes (IFC) schema has emerged as an open standard for encoding BIM semantics, providing object-level access to entities such as IfcSpace, IfcDoor, IfcBuildingStorey, and IfcSensor. While IFC enables semantic modeling, its integration with real-time occupant-level sensing remains technically challenging due to differences in data structure, update frequency, and representation granularity.

Recent research has addressed aspects of BIM–sensor integration from multiple perspectives. Sensor-enabled BIM frameworks using CCTV, BLE beacons, and environmental monitoring devices have been proposed, but are typically evaluated in controlled or simulation-based environments and lack full semantic alignment between IFC entities and sensing stream [10–13]. Studies on IFC extensions and semantic modeling have advanced data representation and ontology design [14–18], but often focus on static scenarios rather than operational real-time response. Agent-based evacuation models and VR/AR-assisted evaluations of underground and transit environments provide insight into behavioral dynamics [19], yet remain largely offline. BIM–IoT and BIM–GIS integration has been explored mainly within facility management contexts rather than time-critical evacuation [20–23].

Recent semantic IoT–BIM routing models offer dynamic path planning capabilities [17,18,24], but most rely on pre-modeled conditions and do not incorporate live crowd sensing or field-level validation.

Overall, the literature reveals three persistent gaps. First, there is a semantic gap between IFC-based building representations and heterogeneous crowd-sensed data streams. Second, many BIM-based evacuation systems exhibit a temporal gap, remaining static or simulation-driven without real-time adaptation. Third, there is an operational gap, as empirical validations in functioning buildings remain limited. Consequently, there is a need for engineering informatics methodologies that explicitly address semantic interoperability, integrate live sensing into routing logic, and demonstrate performance in operational environments.

This study addresses these gaps by proposing an integrated BIM–MCS framework for real-time adaptive evacuation routing. The framework resolves interoperability between IFC-based semantics and live occupant-level sensing and exploits this integration to support multi-criteria, context-aware route computation. Unlike existing BIM-based evacuation models that remain largely simulation-driven, and crowd-sensing systems that operate independently of architectural semantics, this study establishes a semantic-level operational integration between IFC-based BIM and live mobile crowd sensing (MCS). The proposed framework transforms BIM entities into a continuously updated decision graph whose geometry, semantics and weights are governed by real-time occupant behavior and environmental sensing. This results in a digital-twin-like system that performs real-time computational reasoning over buildings, rather than static analysis or offline simulation. To the authors' knowledge, this is the first study to demonstrate a field-validated evacuation system in which IFC semantics, mobile sensing, and multi-criteria routing are unified within a single operational framework.

Building on these contributions, this study further advances the field of Engineering Informatics by showing how IFC-based BIM models can be transformed from static digital representations into operational computational systems for real-time decision support. The proposed framework converts IFC entities into a dynamic decision graph in which building semantics, spatial structure, and functional constraints are continuously updated using live mobile crowd sensing data.

Rather than treating BIM and sensing as separate components, this work establishes semantic interoperability at the data-model level, allowing occupant behavior, hazard indicators, and exit availability to directly influence routing logic. This enables real-time, multi-criteria computational reasoning within the building model rather than offline simulation.

From a digital twin perspective, the framework demonstrates how BIM can be operationalized through continuous state synchronization between physical spaces and their digital counterparts. This integration advances BIM-centered digital twins from passive representations toward active decision-support systems.

The remainder of this paper is organized as follows. Section 2 reviews related work on BIM-enhanced evacuation systems, mobile crowd sensing, and integrated BIM–MCS approaches. Section 3 presents the proposed framework, including system architecture, IFC-based graph construction, sensing integration, and the routing algorithm. Section 4 reports the results from a three-scenario case study. Section 5 discusses implications for engineering informatics, digital twins, and emergency management, as well as limitations and future directions. Section 6 concludes the paper.

2. Background

Emergency evacuation systems are essential for safeguarding human life during crises such as fires, structural failures, and natural disasters. However, conventional evacuation strategies are predominantly static and exhibit limited responsiveness to real-time conditions. They often fail to accommodate evolving hazards or human behavior, which can result in congestion, confusion, and delayed evacuations [25]. Inadequate situational awareness, stemming from insufficient real-time data further, hampers the ability of emergency managers to make informed decisions [26,27]. Although numerous technological interventions have been explored, a significant disconnect remains between research efforts and their application in operational building environments [28,29]. This section reviews the current body of literature in three main areas: BIM-enhanced emergency evacuation systems, mobile crowd sensing and crowdsourced systems, and integrated BIM–MCS approaches.

2.1. Bim-enhanced emergency evacuation systems

BIM has emerged as a vital tool in emergency management due to its ability to provide semantically rich and geometrically accurate digital models of the built environment [30]. Numerous studies have applied BIM for evacuation route simulation, victim localization, and performance visualization. For instance, Li et al. [31] proposed a BIM-centered indoor localization algorithm to support building fire emergency response operations, while Yoon et al. [32] developed a victim localization system to aid first responders. Chen and Chu [33] presented an API-based system combining BIM and in-building rescue routing. Similarly, a study on BIM-enabled Facility Management in nursing homes highlights how BIM models integrate with digital twins and building sensors to support systemized operations and emergency preparedness in real-world settings [34].

Recent work also explores BIM integration with emerging technologies such as Bluetooth [12], VR/AR [35–39], and digital twins [6,7], demonstrating its potential for improving evacuation awareness and interactivity. Kanangkaew et al. [40] developed a real-time BIM–AR fire evacuation system, while Ahn et al. [38] demonstrated AR navigation for indoor emergencies. However, BIM models are often static and lack real-time adaptability unless enhanced by external sensing mechanisms. Kazado et al. [21] highlighted this limitation and advocated for the integration of sensor technologies with BIM to support facility operations during emergencies.

Research on extending the Industry Foundation Classes (IFC) data model has also gained momentum. Studies have focused on extracting geospatial and evacuation-related features [41,42], enabling path planning in 3D indoor environments [43,44], incorporating sensor and occupant data [19,22], and supporting pedestrian simulations [45]. Despite these efforts, the integration of BIM with live sensor data remains a technical and practical challenge in real-world emergency settings.

2.2. Mobile crowd sensing and crowdsourced systems

Mobile Crowd Sensing (MCS) utilizes the sensing capabilities of smartphones and other personal devices to collect real-time data on user movement, environment conditions, and disaster events. This approach is highly scalable and supports decentralized data collection, enabling dynamic and distributed situational awareness during emergencies. Tripathi and Singh [46] proposed a heterogeneous data fusion model using participatory sensing, while Nguyen et al. [47] designed a mobile app for crowdsourced disaster risk data validation. Cicek and Kantarci [27] categorized MCS applications into eight functional areas, including evacuation mapping, danger detection, and situational awareness. Several systems employ location data from smartphones to support post-disaster navigation. Bhattacharjee et al. [48] developed a crowdsensed digital pedestrian mapping tool, and Fajardo and Oppus [49] proposed a mobile logistics app for disaster relief routing. Wan et al. [50] utilized crowdsourced Wi-Fi signal data to enable multi-floor indoor positioning, while Kitazato et al. [51] estimated pedestrian flow using Bluetooth signals. Asiminidis et al. [52] applied BLE sniffing in tunnel evacuations, and Pascacio et al. [53] provided a BLE dataset to support indoor range estimation. Advanced sensing methods such as gyroscope and camera-based input have also been used for 3D mapping. Sadhu et al. [37] introduced Argus, which uses smartphone sensors to generate a 3D reconstruction of disaster zones, and Piscitello et al. [54] presented a Danger System integrating microphone and pedometer data to detect hazards. Leitch et al. [55] developed a crowd sensing system that systematically examines multi-sensor fusion strategies using Wi–Fi, Bluetooth, ultra-wideband, and inertial measurement units on smartphones. Nonetheless, several limitations persist, including short-range sensor constraints, dependency on battery life, app pre-installation assumptions, and privacy concerns in location tracking [56,57]. While these tools offer high responsiveness, they typically lack integration with architectural context or spatial intelligence provided by BIM.

2.3. Combined BIM–MCS approaches

Despite the complementary strengths of Building Information Modeling (BIM) and Mobile Crowd Sensing (MCS), their full integration in evacuation planning systems remains limited. Early efforts explored partial coupling between building models and mobile technologies. For example, Chen and Chu [33] proposed a BIM–smartphone API framework, while Pishdad-Bozorgi et al. [23] examined GIS–IoT–BIM integration for facility-level coordination. More recent studies have incorporated Bluetooth Low Energy data into BIM-based evacuation models, often combined with crowd simulation techniques [12].

Beyond data-level integration, several ontology- and semantics-driven approaches have been proposed to improve interoperability. Deng et al. [18] introduced a BIM-integrated ontology for fire evacuation, while Pang et al. [17] developed a semantic IoT–BIM framework for dynamic path planning. Related works have also explored BIM–GIS and BIM–IoT combinations for hazard-aware evacuation [15,16]. However, these approaches are typically evaluated in experimental or simulation-based settings and lack validation in operational building environments. Similarly, studies incorporating crowd behavior and congestion modeling, such as those by Ji et al. [11] and Deng et al. [10], primarily rely on offline or simulated scenarios. Other frameworks demonstrate BIM-based automation or agent-based evacuation planning [14,58,59], but do not integrate real-time user-level sensing data. Likewise, BIM–IoT solutions focusing on facility management [20] are not designed for time-critical evacuation contexts. Recent semantic IoT-based routing models [17,18,24,60] offer indoor and underground navigation capabilities, yet often depend on pre-modeled conditions and lack real-time feedback from occupants. Blockchain-based crowd-sensing systems have also been proposed for trust management and post-disaster mapping [61], but remain largely conceptual. Overall, existing research tends to fall into two broad categories: BIM-based evacuation models that remain largely static or simulation-driven, and MCS-based systems that provide responsiveness but lack architectural and semantic contextualization. Full semantic interoperability between IFC-based BIM models and live crowd-sensed data streams remains rare, and empirical validation in functioning buildings is still limited. These gaps highlight the need for integrated BIM–MCS frameworks that support real-time, context-aware evacuation routing under operational conditions. In response, this study proposes an IFC-based BIM–MCS framework that tightly couples semantic building representations with live occupant sensing to enable adaptive, multi-criteria evacuation routing in real-world environments.

3. Methods

This section describes the methodological workflow developed for integrating BIM semantics with real-time crowd-sensed information. The overall process follows common structures used in BIM–IoT systems but extends them to support continuous data fusion and adaptive evacuation routing. The method comprises five components: system architecture, BIM–IFC processing, mobile crowd sensing, indoor localization, and multi-weighted route computation.

3.1. System architecture

The system architecture defines how sensing data, building information, and routing computations interact within the proposed BIM–MCS evacuation framework. It is organized into five functional domains that together support real-time data acquisition, semantic processing, and adaptive route generation as shown in Fig. 1. The sensing domain collects raw measurements from occupants' smartphones, including Wi-Fi and BLE signal strengths as well as contextual indicators such as barometric pressure and ambient light levels. Barometric pressure measurements are incorporated as an auxiliary sensing input to support floor-level differentiation and vertical movement detection in multi-story buildings, thereby improving the robustness of Wi-Fi–based indoor localization during evacuation. These data are transmitted to the collection domain through the network domain, which ensures secure and continuous communication between mobile devices and the server.

Within the collection domain, incoming measurements are stored in a unified repository where they are associated with corresponding IFC entities extracted from the BIM model. The processing domain acts as the core computational layer, performing data filtering, feature extraction, semantic alignment, and dynamic updating of the evacuation graph. Continuous updates from MCS and IFC-linked sensors allow the routing engine to adjust edge weights based on changing conditions such as congestion, hazard proximity, and exit accessibility. The application domain provides the interface through which evacuation instructions and updated paths are visualized for occupants or emergency managers. Fig. 2 shows class diagram and Fig. 3 illustrates the integrated workflow connecting BIM semantics, mobile sensing, and the routing engine within the proposed architecture.

3.2. BIM model

The BIM model provides the spatial and semantic foundation of the proposed evacuation framework. To ensure interoperability and robustness across heterogeneous modeling practices, the building model is exported to the Industry Foundation Classes (IFC) format, which enables standardized access to architectural and spatial entities while remaining independent of authoring software.

Relevant IFC entities are extracted to construct a semantically enriched evacuation graph. Spatial hierarchy is defined using IfcBuilding, IfcBuildingStorey, and IfcSpace, while circulation-related elements such as corridors, staircases, and vertical connections are identified through their spatial containment and adjacency relationships. Rather

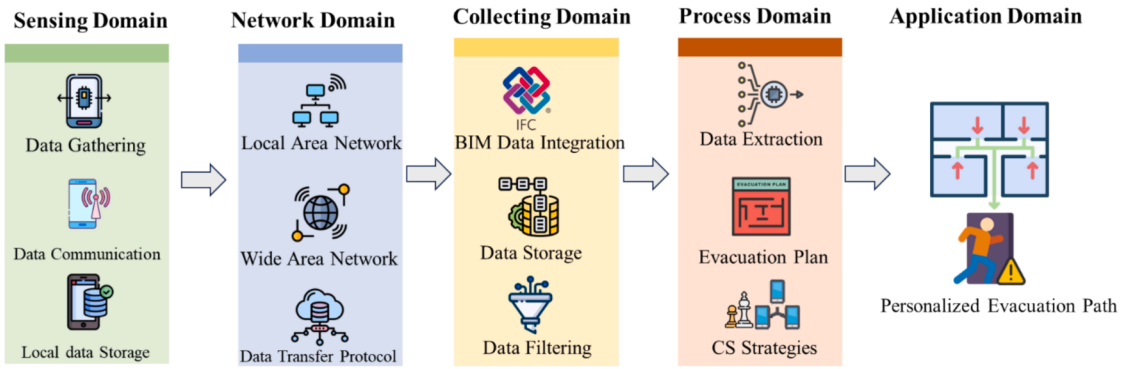


Fig. 1. Proposed integration method.

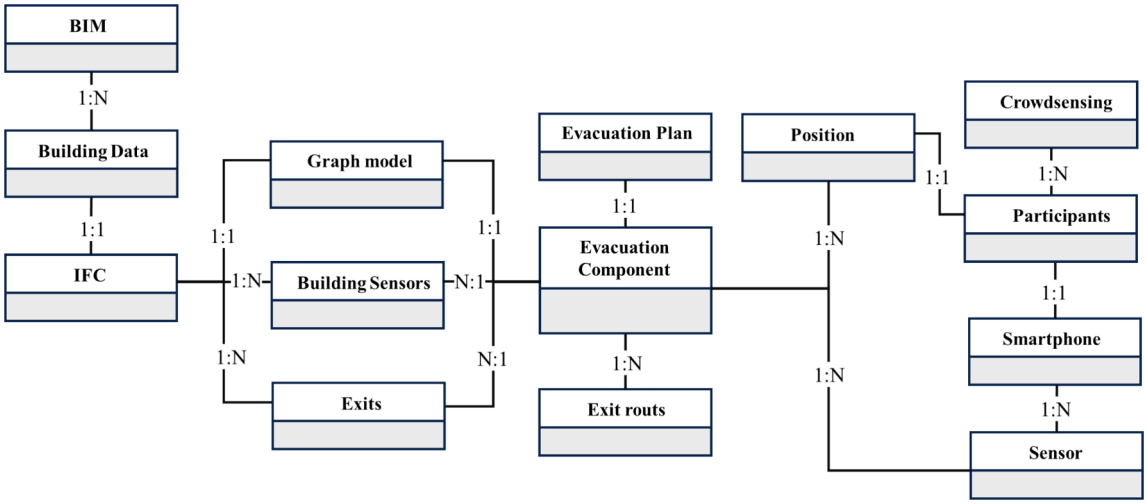


Fig. 2. Class diagram of proposed method.

than relying on a single IFC object type, the framework emphasizes semantic roles (e.g., navigable space, transition, exit) over specific schema classes to support consistent graph construction.

To address heterogeneity in IFC representations, an explicit normalization step is applied during IFC parsing. In IFC-based models, passable connections between spaces may be represented either explicitly by IfcDoor elements or implicitly by wall openings modeled as IfcOpeningElement, depending on authoring conventions and export configurations. In the proposed framework, both representations are abstracted into a unified Transition entity in the evacuation graph. The IFC schema elements used in this semantic extraction process are summarized in Fig. 4.

Transitions are identified primarily through IFC relational semantics rather than object labels. Specifically, IfcRelVoidsElement and IfcRelFillsElement relationships are used to associate openings and doors with their host elements, while space-boundary and spatial-containment relations are employed to determine adjacency between IfcSpace entities. When an IfcDoor object is present, the corresponding transition inherits door-specific attributes such as accessibility status. When only an opening is available, a passable transition is still created using default attributes. This normalization strategy ensures equivalent graph connectivity across different IFC representations and prevents loss of navigable links due to modeling variations.

While IfcSpace entities provide the basis for indoor spatial representation, the term "space" is used differently in the IFC schema and in the proposed graph abstraction. In the proposed abstraction, decision-oriented spaces such as rooms and junctions are represented as nodes, whereas traversal-oriented spaces, including corridors, stairwells, and elevator shafts, are abstracted as transition elements (edges) connecting decision nodes. Environmental conditions such as hazards, congestion, or accessibility constraints primarily affect movement feasibility and are therefore applied to edges rather than nodes.

Based on the extracted and normalized entities, the IFC model is transformed into a graph structure in which nodes represent indoor spaces and edges represent navigable transitions. The resulting graph encodes both geometric connectivity and semantic constraints, including vertical circulation, region-specific accessibility, and exit relationships. This semantically grounded graph serves as the basis for dynamic evacuation routing and enables subsequent integration of real-time crowd-sensed and hazard-related information. Although geometric discretization of walkable spaces is avoided, geometric attributes such as distance and traversal cost are explicitly computed on the derived graph edges (Transitions) after graph abstraction.

3.3. *Mobile crowd sensing*

MCS provides the dynamic component of the proposed framework by supplying real-time information on occupant locations and environmental conditions. Smartphones carried by building occupants continuously collect Wi-Fi and Bluetooth Low Energy (BLE) signal strength measurements for indoor localization, together with contextual indicators such as barometric pressure, ambient light levels, and device-level status information. These measurements capture both spatial movement and environmental variations that are relevant to evacuation behavior.

Barometric pressure measurements are incorporated as an auxiliary

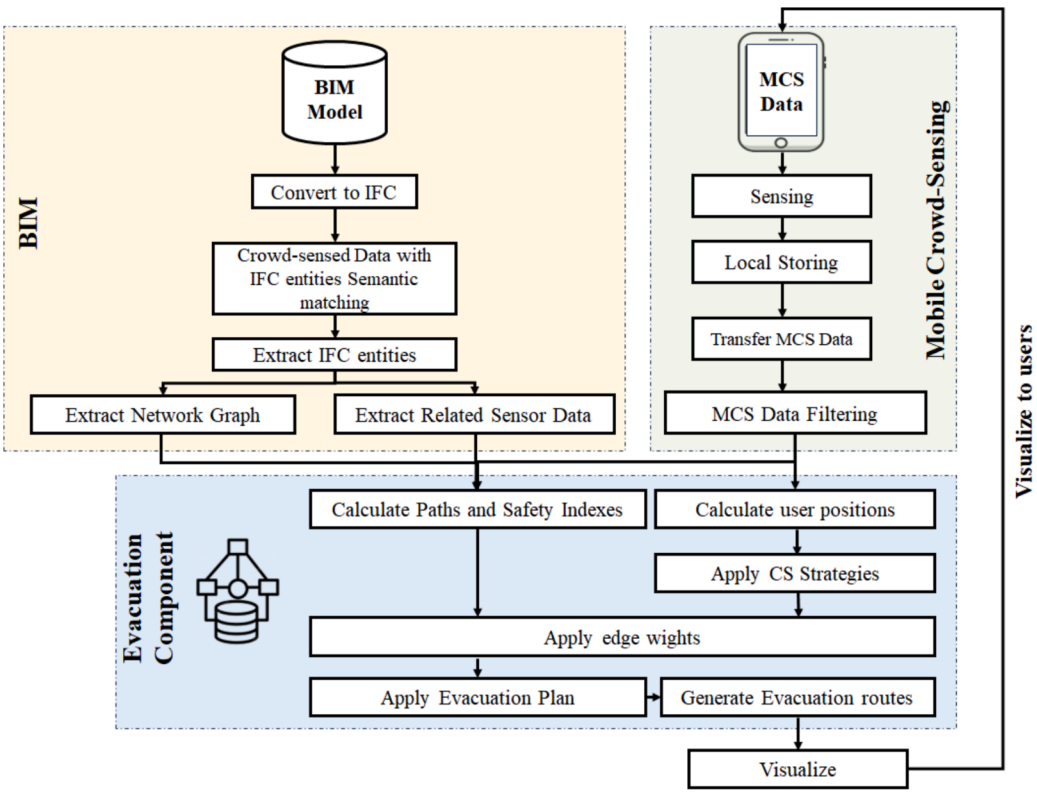


Fig. 3. Overview of the integrated BIM–MCS framework for adaptive evacuation routing.

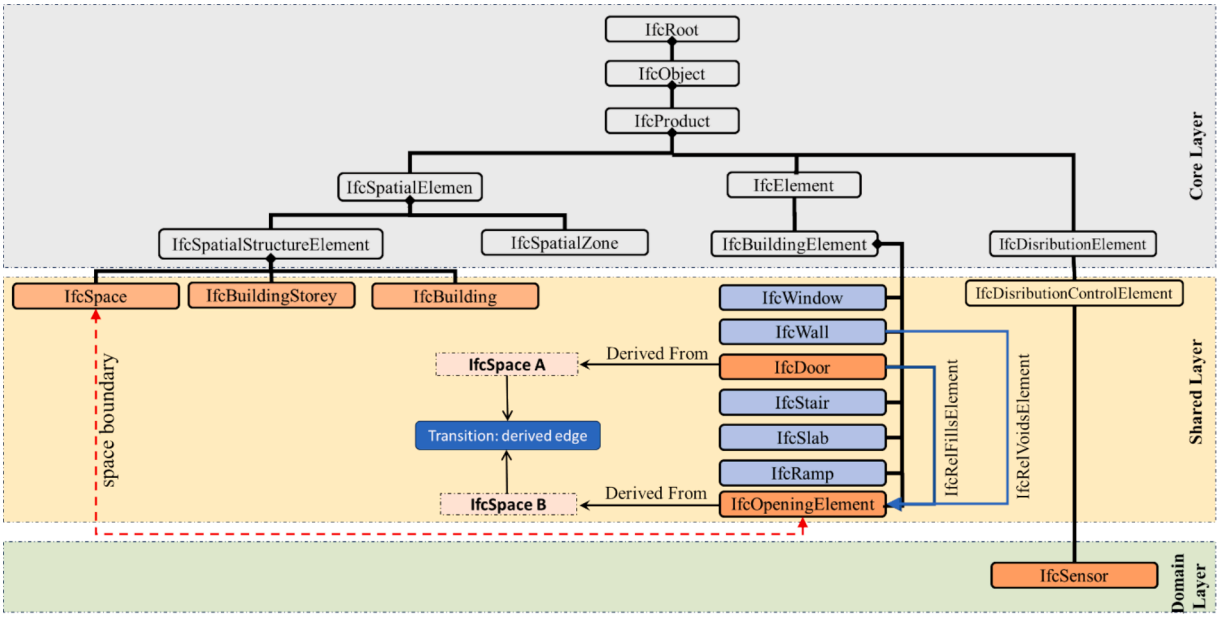


Fig. 4. IFC schema elements and semantic relationships used to derive the evacuation graph. Adjacent IfcSpace entities form graph nodes, while passable connections are inferred through IfcDoor and IfcOpeningElement using IFC relational semantics (IfcRelFillsElement, IfcRelVoidsElement) and space-boundary information. A Transition represents a derived graph edge connecting two adjacent spaces and is not an IFC entity.

sensing input to support floor-level differentiation and vertical movement detection in multi-story buildings. Changes in atmospheric pressure provide lightweight cues for identifying transitions between floors, thereby improving the robustness of Wi-Fi–based indoor localization during evacuation without introducing additional infrastructure requirements. During the initial stage, raw sensing data are stored locally on each device to ensure reliability under unstable network conditions. When connectivity becomes available, the data are transmitted to the central server, where they undergo a preprocessing pipeline that includes redundancy removal, temporal filtering, and quality assessment. The resulting cleaned dataset provides a robust representation of occupant distribution and movement patterns and is subsequently mapped onto the IFC-based spatial graph derived in Section 3.2.

This mapping enables real-time updates of the evacuation model by

aligning user positions with the navigable indoor spaces defined in the BIM representation. In addition to location updates, crowd-sensed information supports the identification of vulnerable occupants, detection of congestion or blocked routes, and interpretation of evolving environmental conditions. As a result, the evacuation model remains responsive to dynamic indoor situations, even when some participants do not actively transmit data.

3.3.1. Evacuation Plan

The Evacuation Plan defines a set of high-level decision rules that govern how evacuation routing is adapted under different emergency conditions. Rather than introducing additional building-specific configurations, the Evacuation Plan operates as a policy layer that translates hazard states and contextual constraints into routing decisions within the evacuation graph.

The questions specified in the Evacuation Plan are designed to capture essential operational considerations relevant to emergency evacuation, such as exit availability, route accessibility, and the permissibility of specific building components (e.g., elevators or restricted zones) under a given hazard scenario. Each question functions as a decision trigger that activates or deactivates corresponding routing constraints, ensuring that evacuation strategies remain consistent with safety regulations and situational requirements.

Hazard information is not directly extracted from BIM geometry. Instead, BIM provides the spatial and semantic context such as space topology, connectivity, and the locations of relevant building elements within which hazard data are interpreted. Actual hazard states are derived from environmental sensors, mobile crowd sensing inputs, and system-level observations (e.g., congestion or route obstruction). These hazard inputs are then mapped onto the evacuation graph by modifying edge weights, accessibility flags, or availability conditions associated with affected transitions.

The Evacuation Plan thus serves as an intermediary decision layer between sensed hazard conditions and graph-based routing computation. By separating policy-level decision logic from building geometry and sensing mechanisms, the framework maintains generality across different building types while enabling scenario-specific evacuation behavior. This design allows the same Evacuation Plan structure to be reused across buildings, with instantiation determined by available BIM semantics and real-time operational data rather than hard-coded configurations.

Based on these policy-level decision rules, Table 1 summarizes the Evacuation Plan items and their effects on evacuation routing.

Table 2 lists the crowd-sensing strategies used in the framework, together with their corresponding system-level prerequisites required for successful execution.

3.4. Evacuation component

The evacuation component constitutes the core analytical unit of the proposed framework, integrating semantic building information from the BIM model with real-time MCS data to generate adaptive evacuation routes. The IFC-derived spatial graph is continuously updated with dynamic edge weights that represent changing indoor conditions, including congestion levels, hazard proximity, accessibility, and exit status. Through these updates, the routing engine maintains an accurate and real-time representation of the building environment during emergencies.

Table 2
Crowd-sensing strategies.

CS strategy	Description	Required system prerequisites
Detect congestion	Identify high-density areas and crowd accumulation using aggregated user movement and location data	Availability of Wi-Fi/BLE signals; sufficient number of participating devices
Identify blocked paths	Infer blocked or obstructed routes based on abnormal movement patterns or sudden flow interruptions	Continuous sensing data; reliable indoor connectivity
Select coordinators / guides	Designate selected occupants as coordinators to assist evacuation based on location and mobility patterns	Active user participation; sufficient battery level and network access
Detect silent users	Infer the presence and approximate locations of occupants who do not actively transmit data	Crowd-density estimation; historical movement patterns; nearby active users
Assess hazard proximity	Estimate proximity to hazardous areas (e.g., smoke-affected zones) using crowd-sensed and environmental data	Environmental sensing inputs; mapping to BIM-based spatial context

3.4.1. Positioning

Accurate occupant localization is essential for computing real-time evacuation routes. The proposed system employs Wi-Fi–based indoor positioning, in which received signal strength indicators (RSSI) are used to estimate the distance between a mobile device and multiple access points. Equation (1) expresses the standard RSSI-to-distance conversion model, where A represents the reference signal strength and n is the environment-specific path-loss exponent. These estimated distances form the input to trilateration or fingerprinting methods, refined through an optimization procedure defined in Equation (2), which minimizes the difference between measured and predicted distances. This approach compensates for multipath interference, device variability, and signal fluctuations common in indoor environments.

A two-stage positioning strategy is used to address cases where some users are unable to transmit sensing data. In the first stage, locations of active users are estimated directly from RSSI-based trilateration. These positions then serve as reference points for the second stage, in which locations of silent users are inferred through proximity relationships derived from the signals of nearby active users. As illustrated in Fig. 5, this strategy ensures a complete spatial distribution of occupants even under inconsistent data availability, enabling the evacuation engine to maintain a coherent occupancy map under real-world conditions.

$$D = d_0 10^{\frac{RSSI-A}{10n}} \tag{1}$$

$$minimize \sum_{k=0}^{n} \left((x - x_i)^2 + (y - y_i)^2 - D_i^2\right)^2 \tag{2}$$

3.4.2. Route Finding

The routing engine extends Dijkstra's algorithm by incorporating

Table 1
Evacuation Plan Items.

Evacuation plan items
Lifts are/are not usable during evacuation: yes ☐ no ☐
Starecases are/ are notusable during evacuation: yes ☐ no ☐
are thereany individuals requiring additional assistance(high risk persons: kids,elderies, disbled persons)? yes ☐ no ☐
individuals selected as coordinator/informer in the evacuation yes ☐ no ☐
extract hidden individuals from mobile crowd sensing data? yes ☐ no ☐
Location of key safety hazards or other related equipment condidered in calculating evacuation routes: yes ☐ no ☐

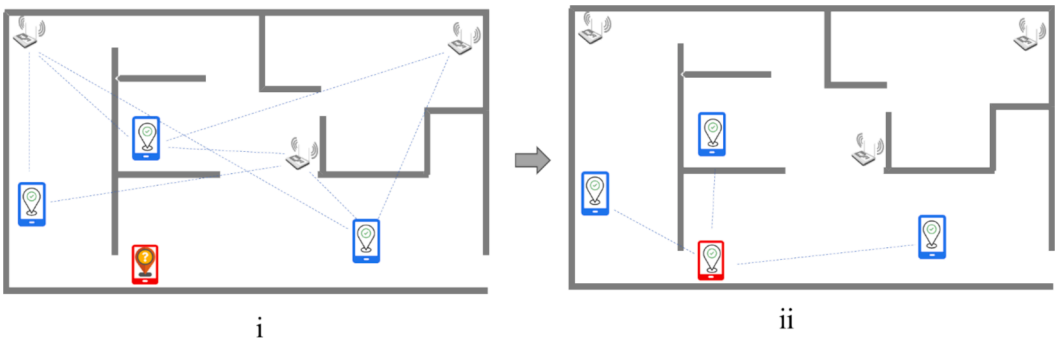


Fig. 5. Two-stage crowdsensing positioning.

four weighted criteria: normalized distance to hazard, travel time, exit availability, and a Route Safety Index (RSI). Distance normalization is defined in Eq. (3), and travel time is computed using Eq. (4) based on edge length and walking speed, which varies with corridor type, vertical circulation, and user characteristics. Exit availability is enforced directly in the cost function by assigning prohibitive penalties to blocked exits.

$$D = \frac{\mathrm{X} - \mathrm{Min}}{\mathrm{Max} - \mathrm{Min}} \tag{3}$$

$$T = \frac{\mathrm{Length}_{edge}}{\mathrm{Speed}_{edge}} \tag{4}$$

The Route Safety Index (RSI) represents an aggregated environmental risk indicator derived from BIM-integrated sensors and Mobile Crowd Sensing (MCS) data, reflecting conditions such as smoke intensity, crowd density, temperature, and hazard proximity. In the present implementation, RSI is formulated primarily based on smoke density due to its dominant role in fire-related evacuation scenarios. Let S denote the normalized smoke intensity along a route segment. RSI is defined in Eq. (5) and constrained to the interval $(0,1]$, where 1 denotes the maximum normalized value and lower values indicate higher environmental risk. Through this formulation, the routing engine prioritizes evacuation paths that minimize exposure to hazardous smoke in addition to geometric distance and travel time.

$$RSI = \frac{1}{1 + \mathrm{S}} \tag{5}$$

The criteria are combined using the weighted cost function defined in Eq. (6). Variable D denotes the normalized distance to the hazard location, T represents the normalized travel time, and E encodes exit availability, with values set to 0 for accessible exits and to a large penalty (e.g., 10^6) for blocked exits to ensure infeasible routes are excluded from selection. RSI refers to the normalized Route Safety Index. The weights w_1, w_2, and w_3 correspond to distance, time, and safety criteria, respectively, and are constrained to sum to one.

$$Cost = (W_1 D + W_2 T + W_3(1 - RSI))(E + 1) \tag{6}$$

Initial weight values were informed by domain expertise and prior research in emergency planning. Weight distributions vary by emergency type: in fire scenarios, higher values were assigned to w_3 to emphasize route safety, while in time-critical events such as gas leaks or structural failures, greater emphasis was placed on travel time by increasing w_2. To formalize this process, weights were derived using the Analytical Hierarchy Process (AHP) through expert-based pairwise comparisons, and the resulting priorities were normalized to reflect scenario-specific evacuation objectives. Weight configurations were further evaluated through sensitivity analysis under multiple emergency scenarios, as reported in Table 3.

The multi-weighted Dijkstra algorithm then computes optimal evacuation routes using the continuously updated cost structure. As new MCS and BIM-linked sensor data are received, edge weights are recalculated and evacuation paths are dynamically updated in near real time. Route updates are performed in an event-driven manner. When new hazard information is received or when existing hazards propagate, the attributes of the affected transitions are updated or temporarily disabled, triggering re-routing from the occupant's most recent valid location rather than global graph reconstruction.

Table 3
Scenario-based weight configurations for the evacuation cost function derived through expert judgment and sensitivity testing within the AHP framework.

Scenario Type	W1 (Distance)	W2 (Time)	W3 (1 − RSI / Safety Priority)
Balanced priorities	0.3	0.35	0.35
Safety-first (e.g., fire)	0.15	0.25	0.6
Time-critical (e.g., gas leak)	0.1	0.6	0.3
Mobility-focused (e.g., elderly evacuation)	0.25	0.5	0.25
Short-distance emphasis	0.5	0.3	0.2

3.5. Implementation Setup

The proposed framework was implemented and evaluated in a four-story academic building belonging to the Faculty of Surveying and Geomatics. The building includes two staircases located on the eastern and western sides and an elevator adjacent to the main corridor. A detailed BIM model was created in Autodesk Revit and exported to IFC to support semantic processing. Fig. 6 shows the reconstructed BIM model, floor layouts, and the placement of smoke sensors installed along the main corridors of each floor. The sensor network provides real-time hazard indicators that are integrated directly into the evacuation graph.

The building's spatial configuration was converted into a graph representation (Fig. 7), where indoor spaces and circulation segments were transformed into nodes and edges, respectively. Each edge was assigned attributes related to distance, capacity, and travel type (e.g., corridor, stair, elevator). Hazard-related properties derived from the sensor network were also embedded into the graph to allow real-time updates of edge weights.

Mobile sensing data were collected using an application developed in Android Studio based on the AWARE framework. The interface (Fig. 8) enables participants to activate sensing modules and transmit data when connectivity is available. Thirteen physical sensors including Wi-Fi, Bluetooth, accelerometer, barometer, ambient light, magnetic field, and proximity and three virtual sensors (battery, network, and processing status) were utilized. Sensed data were stored locally and uploaded to the server during connection intervals, where they were processed and mapped to the IFC-based spatial graph.

User positions were estimated using Wi-Fi fingerprinting as described in Section 3.4. A two-stage crowdsensing approach was used to infer the locations of hidden users who were unable to upload their data. Coordinators or informers were selected according to evacuation-

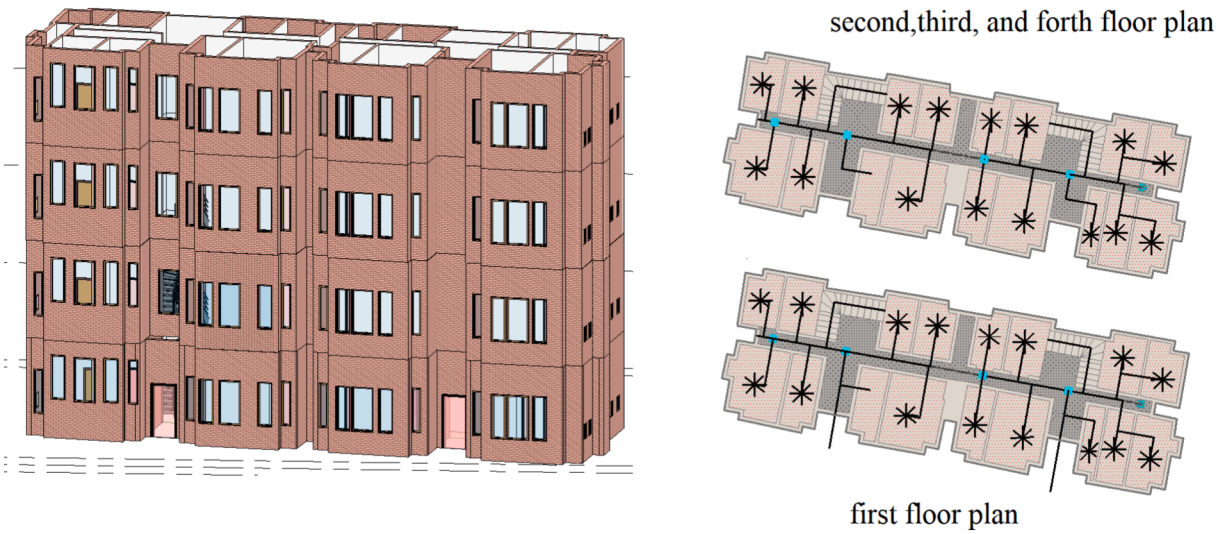


Fig. 6. Case Study: Revit Model (left) and Floor Plans, Rooms, and Smoke Sensors (right).

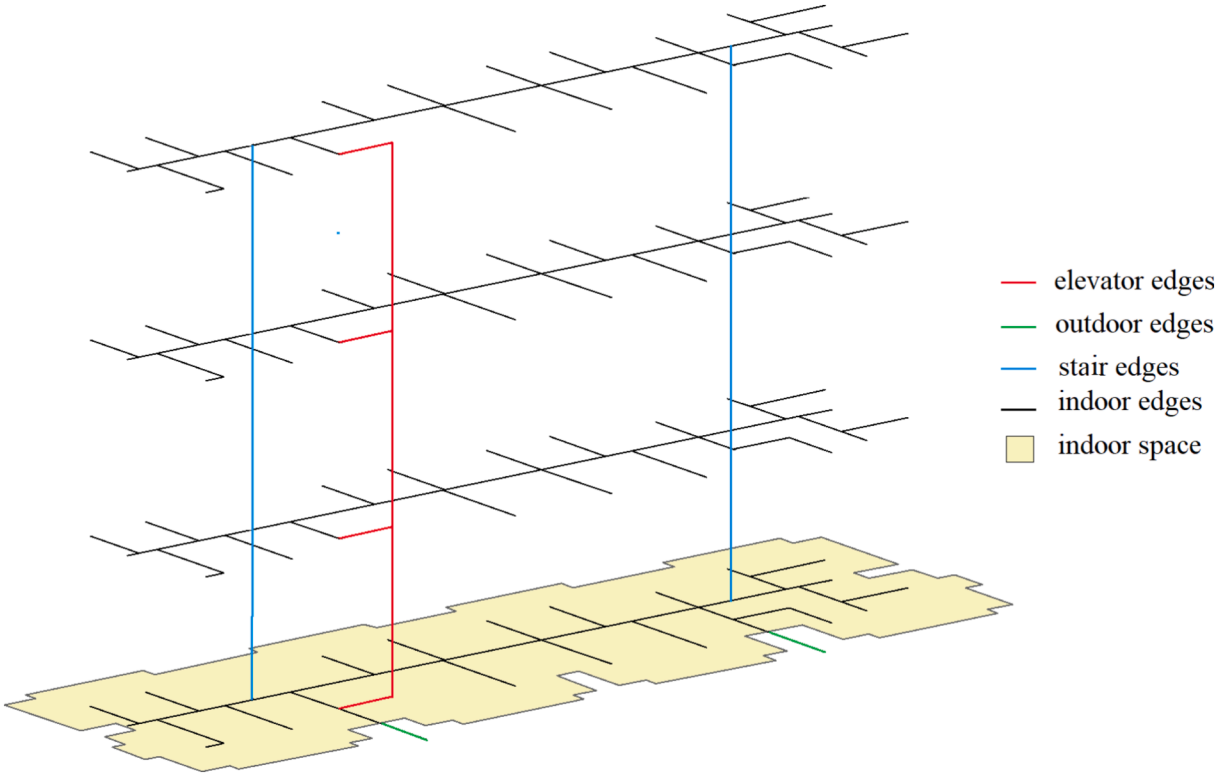


Fig. 7. Extracted graph from the BIM model and associated edge types of the case study.

plan requirements and spatial proximity criteria. The routing engine then generated evacuation paths for each participant based on the dynamic conditions derived from the sensing data and building sensors.

The system was evaluated under three emergency scenarios, summarized in Table 4. Scenario A involved 50 occupants located on the second floor in Room 204 during a fire event, with both the elevator and stairs available. Scenario B included occupants on the western part of the first floor during a structural failure, where the west exit was blocked and coordinators were required. Scenario C simulated an earthquake with occupants distributed across all floors, restricted to stair usage and confronted with severe time constraints. These scenarios were used to assess the adaptability and performance of the routing model under hazard-specific conditions.

4. Results

This section presents the experimental evaluation of the proposed BIM–MCS framework across three emergency scenarios. The findings are reported in terms of route performance, usability, localization precision and accuracy, and evacuation time. The purpose of the analysis is to assess both quantitative system performance and the framework's adaptive behavior under varying hazard and building constraints.

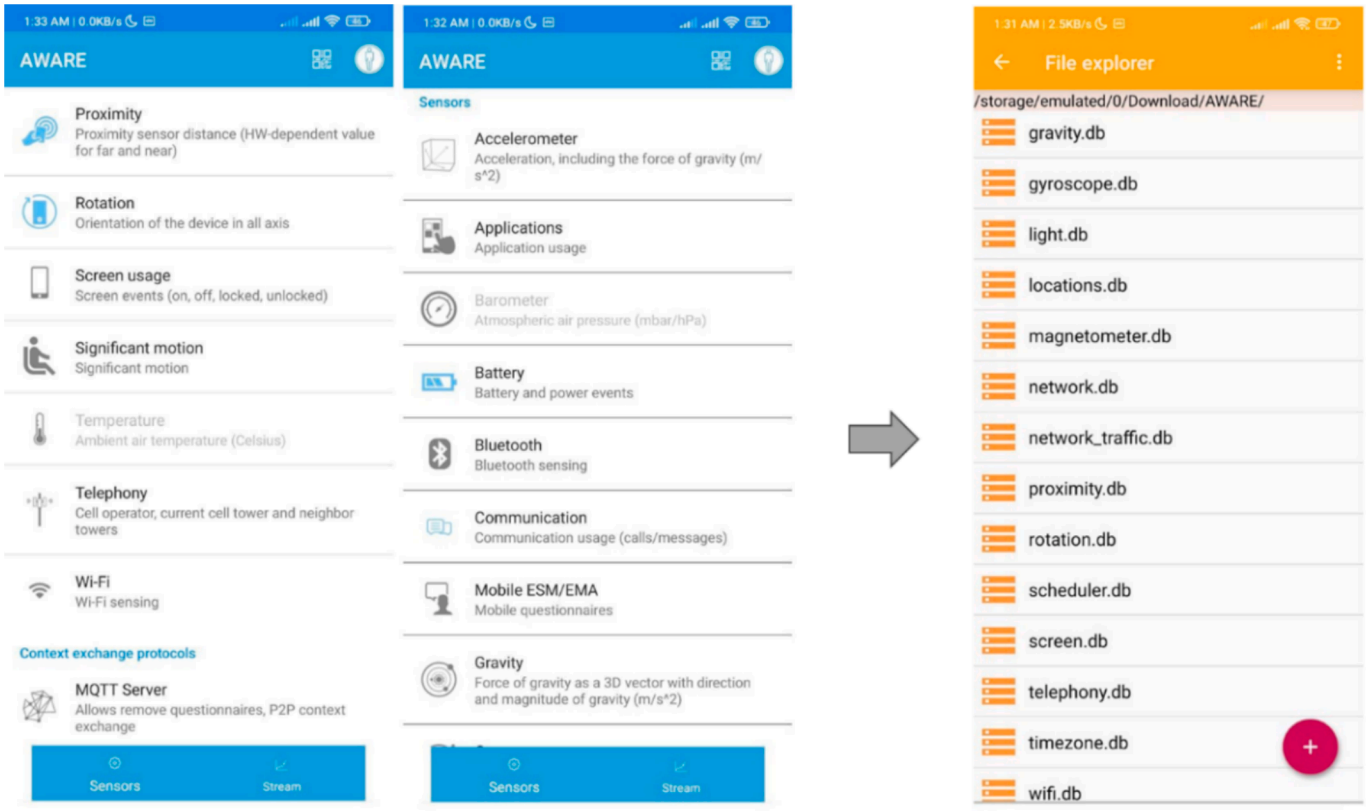


Fig. 8. MCS interface and locally stored crowd sensing data.

Table 4
Summary of evaluation scenarios (*Elevator usage is shown solely for system testing on the complete building graph; elevators are disabled and not used in real fire evacuation guidance.).

Senario	occupant num	location of incident	use of elevator	use of stairs	need of coordinators	incident type	Exit blockage	Time limit
A	50	second floor_room 204	yes*	yes	no	Fire	no	8 minuts
B	50	first floor, western part	yes	only the east stairs	yes	Structural failure	yes	10 min
C	50	whole building	no	yes	yes	Earthquake	yes	2 min

4.1. Route performance metrics

Route performance was assessed using four indicators: average positioning error, average egress time, average route length, and the proportion of in-time versus out-of-time evacuations. Table 5 summarizes the results for all three scenarios.

Scenario A achieved an average positioning error of 3.02 m, while Scenario C exhibited the highest error (3.58 m), reflecting increased localization complexity due to the building-wide distribution of occupants. Despite this, Scenario C produced the shortest average egress time (103 s) and the shortest route length (188 m). These findings indicate that the adaptive routing engine effectively redirected occupants around blocked exits and prioritized feasible short paths even under restrictive conditions. Scenario B yielded the most reliable route completion rates, with 48 in-time evacuations and only 2 out-of-time cases, largely due to the model's successful avoidance of the structurally compromised west exit.

Table 5
Summary of calculated routing and evacuation results.

Senario	Average positioning error(meters)	Average egress/travel time(seconds)	Average route lengths (meters)
A	3.02	111	247
B	3.16	116	221
C	3.58	103	188

Overall, the results demonstrate that the routing engine adapts effectively to spatial and hazard constraints. In scenarios with limited egress options (Scenario B) or building-wide dispersion (Scenario C), the system still identified safe and efficient evacuation paths. Fig. 9 illustrates representative minimum-cost routes generated across the three scenarios. Note that the illustrated route in the fire scenario represents routing on the complete building graph for demonstration purposes. Elevator links are disabled during fire-specific evacuation decision-making and are not used in the final evacuation guidance.

4.2. Usability evaluation

A usability assessment was carried out with all 50 participants to evaluate the Android-based evacuation interface. The interface was rated on navigation clarity, guidance clarity, and responsiveness using a 5-point Likert scale. Average scores were 4.6, 4.2, and 4.4, respectively. High ratings for navigation and responsiveness indicate that the system effectively communicated route updates and that users could follow instructions without confusion. Although more formal usability instruments (e.g., SUS or UEQ) will be considered in future studies, the current results show that the interface provides intuitive and reliable support during evacuation tasks.

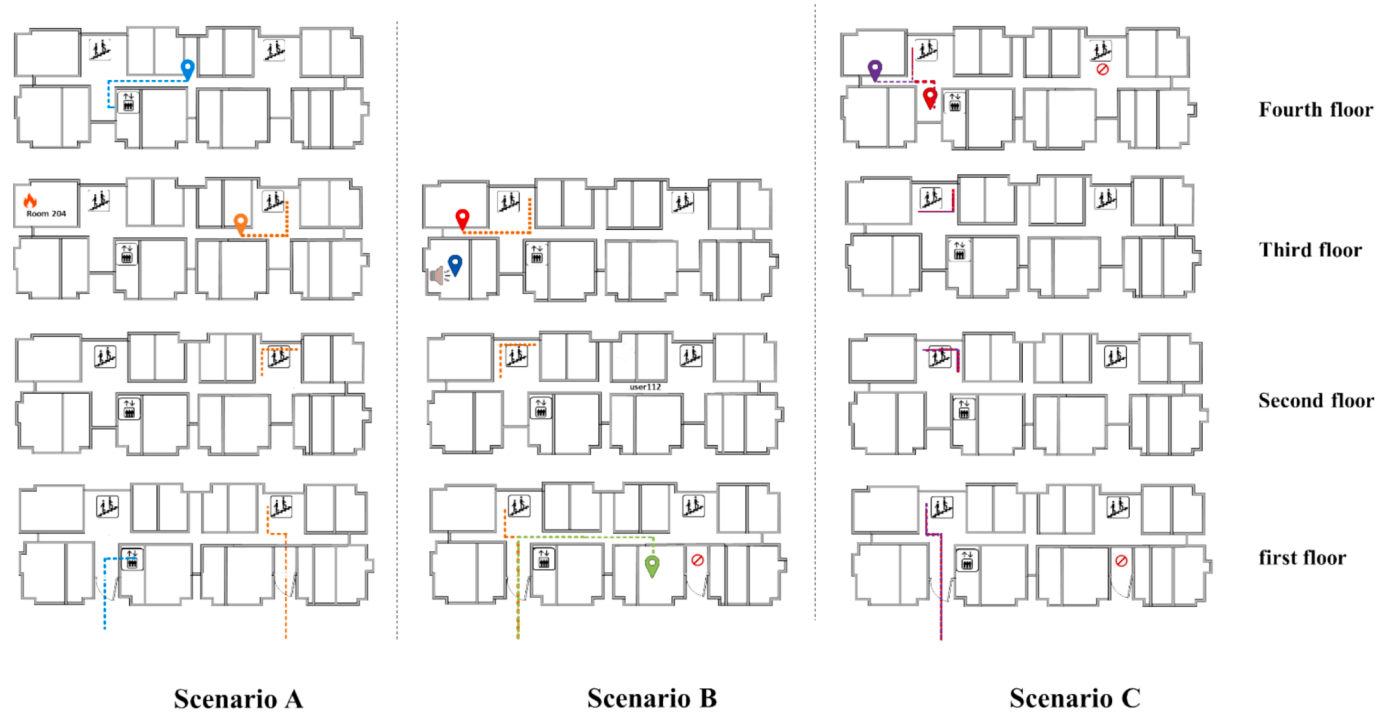


Fig. 9. Sample of visualized minimum cost routes for three evaluation scenarios. The illustrated routes represent routing results on the complete building graph for demonstration and evaluation purposes. In fire-related scenarios, elevator links shown in the visualization represent available building infrastructure and are disabled during evacuation decision-making; final evacuation guidance is therefore restricted to stair-based paths.

4.3. Routing performance and evacuation outcomes

Evaluation is conducted using routing- and system-level metrics derived directly from the outputs of the proposed dynamic evacuation framework. Since the system performs decision-oriented routing under evolving hazards rather than classification or prediction, machine-learning metrics such as accuracy, precision, or confusion-matrix measures are not applicable. Instead, performance is assessed using evacuation-oriented measures that reflect route validity, accessibility to exits, temporal efficiency, and system adaptability.

Route feasibility is characterized using the Route Feasibility Rate (RFR), defined as the proportion of computed routes that remain valid after hazard or obstacle updates:

$$RFR = \frac{N_{feasible}}{N_{Total}}$$

where $N_{feasible}$ denotes the number of routes that continue to lead to an accessible exit without invalidation, and N_{Total} is the total number of computed routes. RFR captures the robustness of routing decisions under dynamic environmental conditions.

Accessibility to exits is described using the Exit Reachability Rate (ERR), which measures the proportion of starting locations that retain at least one valid path to an exit after hazard updates:

$$\mathrm{ERR} = 100 \times \frac{|S_{reachable}|}{|S|}$$

where S represents the set of starting locations (e.g., spaces or origins), and $S_{reachable}$ denotes the subset that remains connected to an exit. ERR serves as an evacuation-oriented proxy for accessibility without requiring agent-level evacuation simulation.

Temporal evacuation efficiency is evaluated using the Average Evacuation Time (AET), computed as:

$$AET = \left(\frac{1}{N}\sum_{i=1}^{N} T_i\right)$$

where T_i is the evacuation time associated with route i, and N denotes the total number of evaluated routes. In addition, evacuation effectiveness under time-critical conditions is assessed using in-time versus out-of-time evacuation outcomes, which distinguish routes that reach an exit within a predefined scenario-specific time limit from those that exceed it.

Based on the experimental results summarized in Table 5 (N = 50 per scenario), the average evacuation times were 111 s, 116 s, and 103 s for Scenarios A, B, and C, respectively. Under the corresponding scenario-specific time limits (8 min, 10 min, and 2 min), the numbers of in-time routes were 47, 48, and 48, while the numbers of out-of-time routes were 3, 2, and 2, respectively. To improve clarity, Table 6 reports time-based evacuation performance metrics derived directly from these results, including in-time and out-of-time rates.

5. Discussion

This study demonstrates that semantic integration between Building Information Modeling (BIM) and Mobile Crowd Sensing (MCS) is a viable mechanism for enabling real-time, data-driven evacuation management in complex indoor environments. The proposed framework moves evacuation planning beyond static path computation toward a continuously updated computational process that reflects changing building conditions and occupant states.

Table 6
Time-based evacuation performance metrics summarizing scenario-specific time limits, average evacuation times, and in-time versus out-of-time evacuation outcomes.

Senario	Time limit (s)	Avg. evacuation time AET(s)	In-time routes	Out-of-time routes	In-time rate (%)
A	480	111	47	3	94
B	600	116	48	2	96
C	120	103	48	2	96

5.1. Semantic coupling of BIM and MCS

A key contribution of this work is the operational alignment between IFC-based building semantics and real-time crowd-sensed data. Conventional BIM-based evacuation models typically rely on precomputed routes or static hazard assumptions, whereas the proposed framework converts the IFC building model into a dynamic graph structure whose nodes and edges are continuously updated using live MCS inputs. This transformation enables indoor spaces, circulation elements, and exits to function as computational entities rather than passive geometric representations.

Through semantic mapping, occupant locations and environmental conditions are bound directly to IFC entities, allowing spatial reasoning and route computation to operate within a building-aware context. This mechanism establishes a persistent link between the physical environment and its digital representation and conforms to the core digital twin requirement of state synchronization between physical and virtual systems.

5.2. Routing behavior under multi-criteria optimization

The evacuation routing formulation extends conventional shortest-path methods by introducing a multi-criteria cost function that accounts for proximity to hazard, travel time, exit availability, and route safety. Unlike distance-only routing strategies, this design treats evacuation as a dynamic optimization problem governed by both spatial and environmental constraints.

The incorporation of a route safety index allows environmental data to influence routing decisions directly. Smoke intensity and crowd density are not handled as secondary indicators but are embedded as control variables within the routing computation. The Analytical Hierarchy Process (AHP) provides a structured approach to weighting parameter selection and reduces subjectivity in cost formulation.

Observed improvements in routing stability indicate that performance gains result from structured multi-criterion integration rather than from dominance of any single factor. Sensitivity analysis confirms that the algorithm adapts to varying hazard profiles without requiring manual retuning, which supports its applicability across multiple emergency contexts.

5.3. Implications for digital twin systems in emergency management

From a digital twin perspective, the proposed system demonstrates how BIM models can evolve from static design artifacts into real-time operational platforms. The integration of live occupancy and environmental sensing enables continuous state updating and route recomputation, forming a closed-loop system between sensing, modeling, and decision execution.

This approach supports the development of smart building systems capable of active intervention rather than passive monitoring. The framework establishes an architectural pattern for BIM-centered digital twins consisting of semantic modeling, sensor integration, dynamic computation, and decision output. This pattern is transferable to other application domains, including energy control, facility automation, and resilience analysis.

For emergency management, the system delivers prescriptive guidance rather than descriptive status updates. Instead of visualizing hazards alone, the system computes and distributes optimized evacuation routes in real time, contributing directly to operational decision-making.

5.4. Reliability and system behavior under constraints

Evaluation results show that evacuation times exhibit lower variance under restrictive conditions. In scenarios with limited exits, routing convergence was more stable due to reduced solution paths. This behavior suggests that constraint-limited environments may exhibit higher algorithmic determinism, leading to predictable outcomes during navigation.

The two-stage crowdsensing localization strategy contributed to maintaining situational awareness even when part of the user population was unable to transmit data. From a system reliability standpoint, this indicates resilience against partial data loss and supports continued operation under degraded communication conditions.

While Wi-Fi–based localization accuracy remains bounded by environmental factors, the confinement of user positions to IFC-defined spaces mitigates uncertainty by constraining feasible locations to architecturally valid zones.

5.5. Limitations and future implications

Several limitations warrant attention. The case study involved a single mid-sized academic building, and further validation across diverse building types is required. Although the experimental validation was conducted in a single academic building, the architecture is independent of building type and relies exclusively on IFC and standard wireless protocols, enabling direct transferability. Wi-Fi localization error remains non-negligible and should be addressed through hybrid positioning strategies incorporating BLE, inertial sensors, or Ultra-Wideband (UWB).

Ethical and privacy challenges associated with mobile crowd sensing must also be acknowledged. Real-time occupant monitoring requires strict data governance mechanisms including anonymization, encryption, and transparency of data usage policies. These considerations are central to sustainable deployment of digital twin technologies in occupied environments. In buildings with highly articulated or irregular circulation geometries, increased spatial complexity primarily results in a larger number of spatial partitions and transitions, yielding a denser evacuation graph rather than requiring methodological changes.

Future work will focus on extending the framework through predictive hazard modeling, sensor fusion strategies, and privacy-preserving data architectures. These developments are necessary to support scalable digital twin deployment in highly occupied and infrastructure-intensive facilities.

6. Conclusion

This paper presented an IFC-based integration framework that tightly couples Building Information Modeling (BIM) with mobile crowd sensing (MCS) to enable dynamic, context-aware evacuation routing in complex buildings. By transforming IFC models into a semantically enriched decision graph continuously updated with crowd-sensed information, the proposed approach moves beyond static BIM-based evacuation models toward operational, real-time decision support.

The main contributions of this work can be summarized as follows. First, a semantic interoperability mechanism was developed to systematically link IFC building entities with live sensing data, enabling dynamic annotation of spaces, exits, and sensors. Second, a graph-based evacuation model was formulated in which routing decisions adapt to evolving hazards, crowd conditions, and exit availability using a multi-criteria shortest-path strategy. Third, a two-stage crowdsensing approach was incorporated to estimate occupant positions by accounting for both active and silent users. Finally, the framework was implemented and evaluated in a real multi-story academic building under multiple emergency scenarios, demonstrating improved evacuation effectiveness under scenario-specific temporal constraints.

Compared to existing BIM-based evacuation approaches, which often rely on static assumptions or precomputed routes, the proposed framework emphasizes continuous data integration and adaptive routing behavior. This positions the framework closer to digital-twin–like operation, where building models are not only descriptive but also actively support real-time operational decision-making during emergencies.

Several limitations of the current study should be acknowledged. The experimental evaluation was conducted in a single building, and the number of emergency scenarios was limited. In addition, while routing adaptability was analyzed through time-based outcomes, detailed agent-level behavioral modeling and large-scale multi-building validation were beyond the scope of this work.

Future research will focus on extending the framework to larger and more diverse building typologies, incorporating richer behavioral and compliance models for evacuees, and integrating additional real-time data sources such as IoT-based environmental sensors. These directions aim to further strengthen the role of BIM-based digital twins as practical decision-support systems for emergency management.

CRediT authorship contribution statement

Mohsen Rashidian: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Mohammad Reza Malek:** Writing – review & editing, Validation, Supervision, Software, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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